Chapter 09 Solid and Fluids

States of Matter

- Solid
- Liquid
- Gas
- Plasma

Solids

- Definite Volume, shape
- Molecules are held in specific locations by electrical forces
- Vibrate about equilibrium positions

Crystalline Solid

- Atoms have an ordered structure. (long-range order)

Amorphous Solid

- Atoms are arranged almost randomly. (Short-range order)

Unit Cell Theory

- Every crystal has a smallest unit cell with a specific shape
- Large crystal are formed by stacking numerous unit cell together

The basic unit that characterizes a crystal structure is called a unit cell. It is a parallelepiped. Lengths of sides and angles between them are used to express unit cell parameter.

X-ray

Electromagnetic waves with in specific wavelength. $(0.01 - 100\mathrm{ANG})$

When thermal electrons accelerated by high voltage bombard on mental target in a vacuum, X-rays are produced.

Liquid

- Definite volume but no definite shape.
- Higher temperature than solid

- Molecules wander through the liquid in a random fashion(The force are not strong enough to keep the molecules in a fixed position)

Gas

- No definite volume and shape
- Molecules are in constant random motion. (Only weak forces on each other)
- Average distance between molecules is larger to the size of the molecules.

Plasma

- Matter heated to a very high temperature.
- Many electrons are freed from the nucleus (ionization)
- A collection of free, electrically charged ions
- Often inside stars and in space.

Deformation of Solids

All objects are deformable by external forces. When the forces are removed, the object tends to its original shape(elastic behavior).

Elastic Properties

- Stress is the force per unit area causing the deformation. ($\text{Stress} = \text{Force}/\text{Area}$)
- Strain is a measure of the amount of deformation. ($\text{Strain} = \text{Change}/\text{Original}$)
- The elastic modulus is the constant of proportionality between stress and strain.

Elastic Modulus

A material with a large elastic modulus is very stiff and difficult to deform.

$$\text{Stress} = \text{Elastic modulus} \times \text{strain}$$

Young's Modulus: Elasticity in Length

Tensile stress is the ratio of the external force to the cross-sectional area.

$$\frac{F}{A} = Y\frac{\Delta L}{L_0}$$

The Young's modulus applies to both tension or compression.
If exceed the elastic limit of the material

- No longer directly proportional

- Cannot return to its original length

Shear Modulus: Elasticity of Shape

If the force are parallel to one of the object's faces, the stress is called a shear stress. The shear strain is the ratio of the horizontal displacement and the height of the object.

$$\frac{F}{A} = S\frac{\Delta x}{h}$$

Bulk Modulus

Volume stress ΔP is the ratio of the force to its surface area. Volume strain is the ratio of the change in volume to the original volume.

$$\Delta P = -B\frac{\Delta V}{V}$$

The negative sign expresses that an increase in pressure will produce a decease in volume. (B is always positive) The compressibility is the reciprocal of the bulk modulus.

Notes on Modulus

- Solid have Young's, Bulk, and Shear
- Liquid only have bulk.

Density

The mass per unit volume

$$\rho = \frac{m}{V}$$

Density of most liquids and solids are affected by temperature and pressure slightly but greatly for gases'.

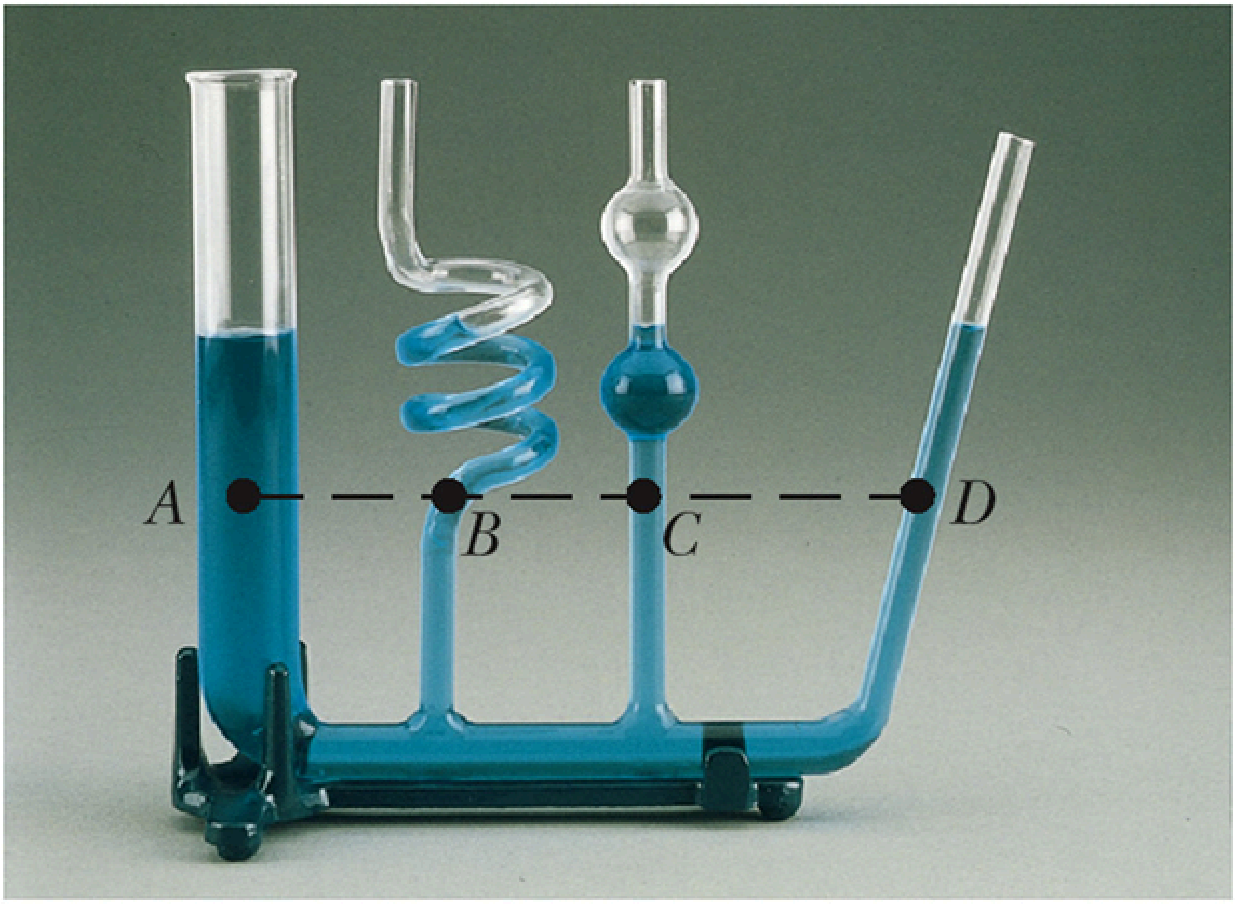
Pressure

The force per area.

$$P = \frac{F}{A}$$

Pressure and Depth

All the points at the same depth has the same pressure. (not depend on the shape of the container)

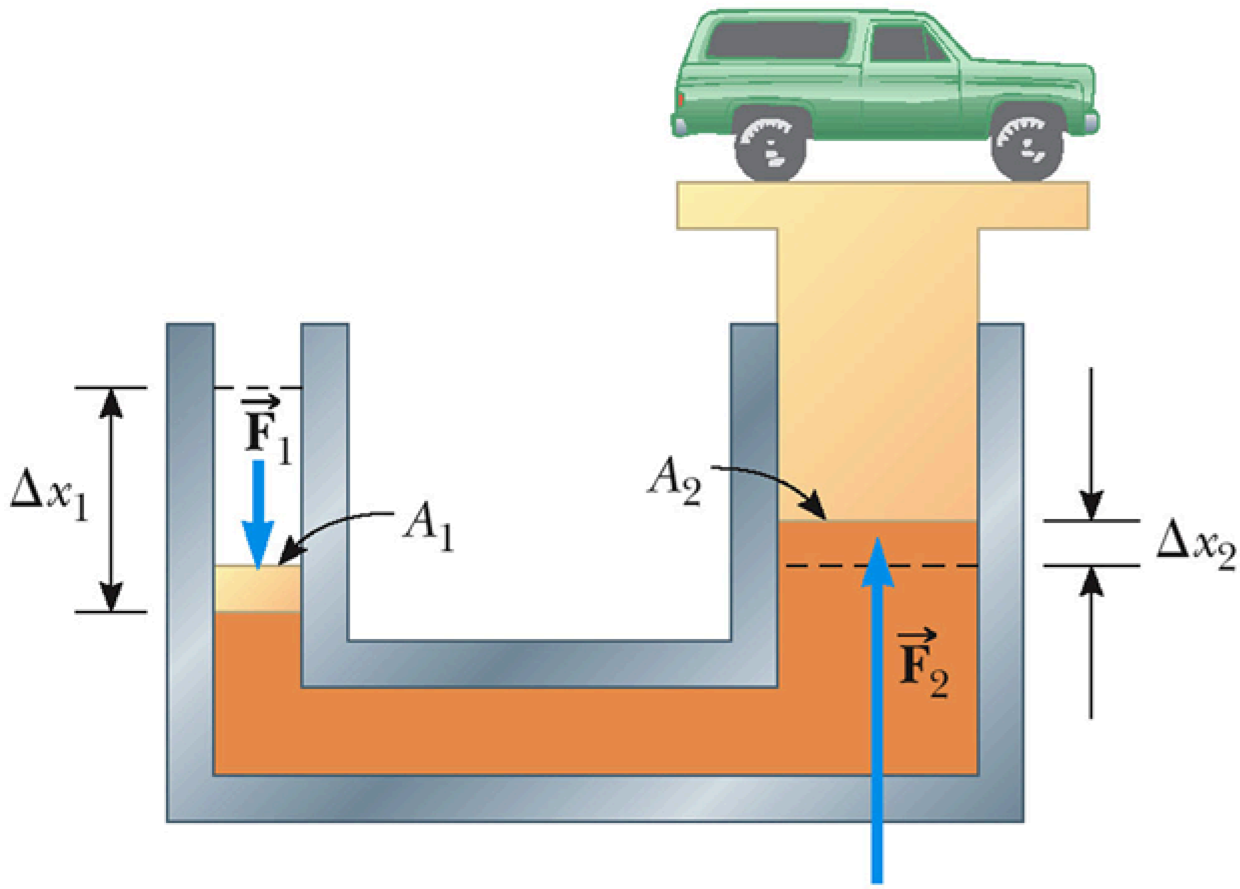


$$p = p_0 + \rho g h$$

where p_0 is normal atmospheric pressure.

Pascal's Principle

A change in pressure applied to an enclosed fluid is transmitted undiminished to every point of the fluid and to the walls of the containers.



$$P = \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

And the work done by input and output pistons is the same. (More distance of input)

Absolute vs. Gauge Pression

- The pressure $P_0 + \rho g h = P$ is called the absolute pressure
- $P - P_0 = \rho g h$ is the gauge pressure.

Archimedes' Principle

The force buoying up an object equals to the weight of the fluid displaced by the object.

Buoyant Force

The upward force is called buoyant force, caused by the difference between the top and the bottom of the object.

$$F_B = \rho_{\text{fluid}} g V_{\text{fluid}} = W_{\text{fluid}}$$

Fluid in Motions: Streamlines

Streamline flow:
Every particle that passes a particular point moves exactly along the smooth path followed by particles that passed the point earlier. (also called laminar flow)
Different streamlines cannot cross each other. And it represents the direction of fluid velocity at that point.

Fluids in Motion: Turbulent Flow

The flow becomes irregular, exceeds a certain velocity.

Characteristic of an Ideal Fluid

- The fluid is nonviscous(no internal friction between adjacent layers)
- The fluid is incompressible(density is constant)
- The fluid motion is steady. (Velocity, density and pressure not change in time)
- The fluid moves without turbulence. (no eddy currents, have zero angular velocity about its center)

Equation of Continuity

$$A_1v_1 = A_2v_2$$

The product of the cross-sectional area of a pipe and the fluid speed is a constant. And Av is called the flow rate.

Bernoulli's Equation

The sum of the pressure, kinetic energy per unit volume and the potential energy per unit volume has the same value at all points along a streamline.

$$P + \frac{1}{2}\rho v^2 + \rho g y = \text{constant}$$